# Systems Architecture

# Decimal Numbers



## Decimal Representation

### Fixed point

(*Fractions*)

### Floating Point

(*Scientific Notation*)



## The Binary Fraction

| $2^2$ | $2^1$ | $2^0 \bullet 2^{-1}$ | $2^{-2}$ | $2^{-3}$ | $2^{-4}$ | $2^{-5}$ | $2^{-6}$ | $2^{-7}$ |
|---|---|---|---|---|---|---|---|---|
| 4 | 2 | $1 \bullet \frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{16}$ | $\frac{1}{32}$ | $\frac{1}{64}$ | $\frac{1}{128}$ |
| 4 | 2 | $1 \bullet \frac{64}{128}$ | $\frac{32}{128}$ | $\frac{16}{128}$ | $\frac{8}{128}$ | $\frac{4}{128}$ | $\frac{2}{128}$ | $\frac{1}{128}$ |
| 0 | 1 | $1 \bullet 0$ | 0 | 1 | 0 | 0 | 1 | 0 |

$$2 + 1 + \frac{16}{128} + \frac{2}{128} = 3 + \frac{16+2}{128} = 3 + \frac{18}{128}$$

$$011 \cdot 0010010_2 = 3 \cdot 140625_{10}$$

## Fractions: Dec to Bin

$$0{\cdot}6875 \times 2 = \mathbf{1}{\cdot}375$$
$$0{\cdot}375 \times 2 = \mathbf{0}{\cdot}75$$
$$0{\cdot}75 \times 2 = \mathbf{1}{\cdot}5$$
$$0{\cdot}5 \times 2 = \mathbf{1}{\cdot}0$$
$$0{\cdot}0 \times 2 = \text{End}$$

$$0{\cdot}6875_{10} = 0{\cdot}1011_2$$



## Fractions: Accuracy

- $0{\cdot}1 \times 2 = \mathbf{0}{\cdot}2$ $\quad \frac{1}{2} \quad \frac{256}{512}$
- $0{\cdot}2 \times 2 = \mathbf{0}{\cdot}4$ $\quad \frac{1}{4} \quad \frac{128}{512}$
- $0{\cdot}4 \times 2 = \mathbf{0}{\cdot}8$ $\quad \frac{1}{8} \quad \frac{64}{512}$
- $0{\cdot}8 \times 2 = \mathbf{1}{\cdot}6$ $\quad \frac{1}{16} \quad \frac{32}{512}$
- $0{\cdot}6 \times 2 = \mathbf{1}{\cdot}2$ $\quad \frac{1}{32} \quad \frac{16}{512}$
- $0{\cdot}2 \times 2 = \mathbf{0}{\cdot}4$ $\quad \frac{1}{64} \quad \frac{8}{512}$
- $0{\cdot}4 \times 2 = \mathbf{0}{\cdot}8$ $\quad \frac{1}{128} \quad \frac{4}{512}$
- $0{\cdot}8 \times 2 = \mathbf{1}{\cdot}6$ $\quad \frac{1}{258} \quad \frac{2}{512}$
- $0{\cdot}6 \times 2 = \mathbf{1}{\cdot}2$ $\quad \frac{1}{512} \quad \frac{1}{512}$
- $0{\cdot}2 \times 2 = \ldots$

$$0{\cdot}1_{10} = 0{\cdot}0001\dot{1}0\dot{0}1\dot{1}_2$$
$$= \frac{32}{512} + \frac{16}{512} + \frac{2}{512} + \frac{1}{512}$$
$$= \frac{32+16+2+1}{512}$$
$$= \frac{51}{512}$$
$$= 0{\cdot}099609375_{10}$$



## Problems with Fixed Point

- Range
  - $0{\cdot}199 \times 10^{34}$ g — Mass of the Sun
  - $0{\cdot}910956 \times 10^{-27}$ g — Mass of an Electron
- Precision (or exactness)

$$\pi = \begin{cases} 3{\cdot}140625 & \text{stored} \\ 3{\cdot}1415926 & \text{actual} \\ 3{\cdot}142 & \text{rounded} \end{cases}$$

- Accuracy

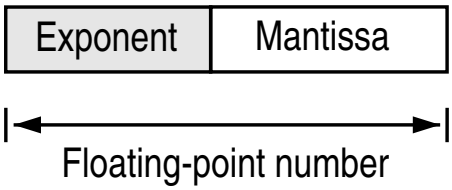
  $0{\cdot}1 \neq 0{\cdot}1$

## Floating Point

$$123{\cdot}45 = 0{\cdot}12345 \times 10^3$$

$$1010{\cdot}1 = 0{\cdot}10101 \times 2^4$$

$$mantissa \times base^{exponent}$$






## Normalization

- Positive numbers (mantissa)
  Alter exponent until mantissa starts $1{\cdot}x$:
  $00{\cdot}011\ldots \times 2^n \Rightarrow 1{\cdot}1\ldots \times 2^{n-2}$
  $10{\cdot}100\ldots \times 2^n \Rightarrow 1{\cdot}01\ldots \times 2^{n+1}$
- Zero is a special case
- Negative numbers (twos complement mantissa)
  Alter exponent until mantissa starts $1{\cdot}x$
- After Normalization mantissa ($m$) is:
  $-1\frac{1}{2} \le m < -1 \qquad m = 0 \qquad 1 \le m < 1\frac{1}{2}$

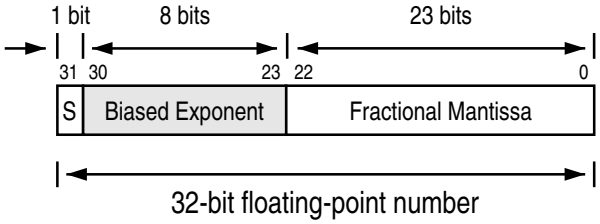


## Alternative representations

- Hidden Bit / Normalisation
  Don't store the leading "$1{\cdot}$" of Normalised number, extending (doubling) precision
- Store mantissa as sign and magnitude
  $0{\cdot}00\ldots00$ to $0{\cdot}11\ldots11$ for Positive
  $1{\cdot}00\ldots00$ to $1{\cdot}11\ldots11$ for Negative
- Biased Exponents (excess)
  Add a (negative) constant to exponent
  $0{\cdot}0$ represented as $0{\cdot}00\ldots00 \times 2^0$
  Exponent treated as unsigned binary integer

## IEEE Floating Point (Single)

$$X = -1^S \times 2^{E-B} \times 1.F$$




| | | | |
|---|---|---|---|
| $S$ | Sign | 0 = +ve, 1 = −ve | $+1{\cdot}0 \times 2^{-127}$ = $0$ |
| $E$ | Exponent | 0...255 | $\pm 1{\cdot}0 \times 2^{128}$ = $\pm\infty$ |
| $B$ | Bias | 127 | $\pm 1{\cdot}x \times 2^{128}$ = NaN |
| $F$ | Fraction | Normalized Mantissa ($1.F$) | (*Not a Number*) |



## IEEE Floating Point Example

$-363{\cdot}375_{10}$ = $-101101011{\cdot}011_2$

= $-1{\cdot}01101011011_2 \times 2^8$

S = −ve = 1

E = $8_{10} + 127_{10} = 135_{10} = 10000111_2$

F = 01 101 011 011 000 000 000 00

| S | E | F |
|---|---|---|
| 1 | 10000111 | 01101011011000000000000 |

1100 0011 1011 0101 1011 0000 0000 $0000_2$ = C3B5B000$_{16}$



## Floating Point Arithmetic

1. Identify number with smaller exponent
2. Make smaller exponent equal to larger
3. Add/Subtract mantissas
4. Normalize the result

- Truncation (move towards $-\infty$)
  Ignore lost bits
  Also known as floored

- Rounding (move towards $0$)
  Add lost bits back into result

# Advantages of Floating Point

- IEEE Standard
- Range
  Defined by size of exponent
  And the value of the bias / excess
- Precision
  Defined by size of mantissa
- Specialist Hardware
  Floating Point Unit / Co-processor